

AS5100 Mini Project

Team Skunk Works - Group 11

Abhijeet Mangela	AE25M034
Manav Agarwal	AE25M035
DS Vivek	AE25M018
Mahesh Kumar S	AE23B019
Soumil BM	AE23B104

UAV CAD Model



Figure 1: Isometric View of the UAV

S. No.	Primary Specification	Value
1	Cruise Velocity	22 m/s
2	Stall Speed	14 m/s
3	Endurance	120 min
4	Absolute Ceiling	4000 m above MSL
5	Payload Weight	2.1 kg
6	Max height from ground	200 m
7	Range	83 km

(a)

S. No.	Derived Specification	Value
1	MTOW	6.9 kg
2	Battery weight	1.53 kg
3	Empty weight	3.27 kg
4	Wing span	2 m
5	Fuselage length	1.09 m
6	Horizontal tail span	0.77 m
7	Vertical Tail span	0.26 m

(b)

1 Mission Statement

Design and develop a fixed-wing Unmanned Aerial Vehicle (UAV) with a maximum wingspan of 2 m capable of safely, reliably, and efficiently delivering a 1 kg emergency medical payload over a maximum range of 80 km.

2 Mission Profile

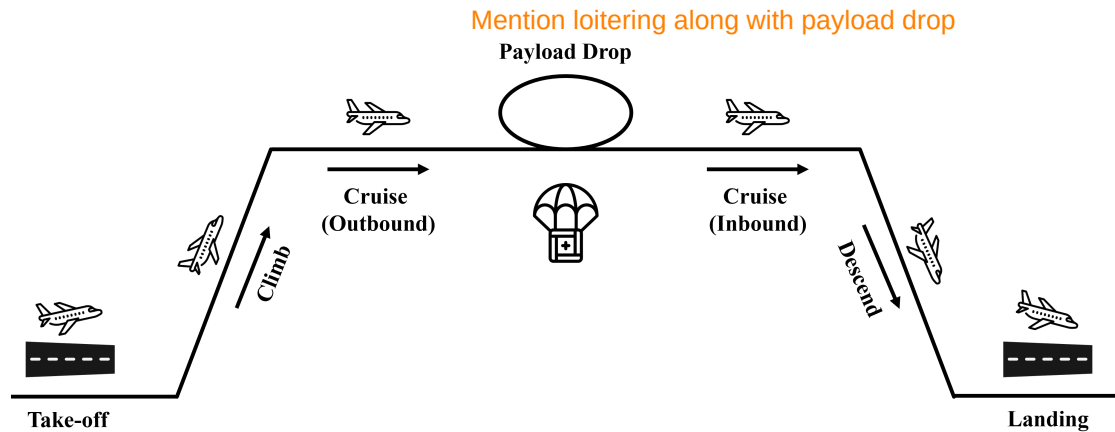


Figure 2: Mission Profile

Why AGL-Above ground level?

S.No	Mission Phase (m/s)	Velocity (m/s)	Segment AGL (km)	Altitude (m)
1	Take-off	-	-	0
2	Climb	20	1.42	0-200
3	Outbound Cruise	22	40 40 Km?	200
4	Payload Drop	22	-	200
5	Loiter (Turn)	22	-	200
6	Inbound Cruise	22	40	200
7	Descent	20	1.42	200-0
8	Landing	-	-	0

Table 2: Mission Profile Parameters

Brief description of mission profile is needed

3 Weight Budget

3.1 Payload Weight Breakdown

Component	Weight (kg)
Variable Drop Payload	1.2
Flight Controller and Telemetry	0.5
Camera and Gimbal (Stabilization & Targetting)	0.2
Electronic Speed Controllers (ESC - 2 units)	0.2
GPS Module	0.04
Total Design Payload	2.14 $\approx$ 2.1 kg

Table 3: Total Payload Weight

3.2 Weight Breakdown

Payload is mentioned as 1Kg in mission statement, whats the additional 200 gm?

Max Take-off Weight	6.9 kg
Battery Mass	1.5 kg
Total Payload	2.1 kg
Empty Weight	3.2 kg

Adds upto 6.8 Kg not 6.9 Kg
Fraction/percentage is missing?

Table 4: Total UAV Weight Breakdown

4 Wing Sizing

4.1 Airfoil Parameters

The Selig S7055 was chosen for the wing due to its performance at low Reynolds numbers flow and the XFLR5 analysis showed that the Airfoil achieves high Lift to Drag ratio within the operational Angle of Attack range.

Dont use Pascal Case in middle of the sentences

Property	Value
Maximum Thickness	10.5 %
Maximum Thickness Location	31.74 %
Camber	3.55 %
$C_{l_{max}}$	1.15
AoA at $C_{l_{max}}$	8°
$(\frac{C_l}{C_d})_{max}$	100
AoA at $(\frac{C_l}{C_d})_{max}$	5.2°

Cd_min, Cm, Cl at L/D max missing

Table 5: Airfoil Parameters

4.2 Wing Parameters

We have chosen a High Wing configuration which provides Roll Stability and Ground clearance. The wing has no dihedral and sweep angle; the planform chosen is rectangular with zero wing twist and $AR = 7.53$. No high-lift devices are used.

Property	Value
Wing Span	2 m
Surface Area	$0.53m^2$
Mean Aerodynamic Chord	0.265 m
$C_{L_{max}}$	1.05
$(\frac{C_L}{C_D})_{max}$	28

Having params like dihedral, taper ratio, twist,etc in table would have been better

Table 6: Wing Parameters

4.3 Control Surface Sizing - Aileron

Property	Value
Aileron Span	0.5 m
Aileron Surface Area	$0.04m^2$
Aileron Chord	0.07 m

Location of aileron?

Table 7: Aileron Sizing

5 Tail Parameters

5.1 Tail Configuration

The Tail configuration chosen is T-Tail to improve the Horizontal tail efficiency by placing it away from the wake of the flow past the Wing

5.2 Tail Airfoil

The horizontal and vertical Tails use NACA0012, a symmetric airfoil to produce symmetric lift with positive and negative AoA.

5.3 Horizontal & Vertical Tail Parameters

Airfoil cross sections?

Parameters	Value
Tail Area	0.132 m ²
Volume Ratio	0.66
Arm Length	0.70 m
Aspect Ratio	4.5
Span	0.773 m
Mean Aerodynamic Chord	0.172 m
Tail Setting Angle	-0.31°

How was tail setting angle calculated?

Table 8: Horizontal Tail Parameters

Parameters	Value
Tail Area	0.047 m ²
Arm Length	0.70 m
Aspect Ratio	1.5
Span	0.268 m
Mean Aerodynamic Chord	0.178 m
Setting Angle	0°

Volume ratio?

Table 9: Vertical Tail Parameters

5.4 Tail Control Surfaces

Property	Value
Rudder Span	0.23 m
Rudder Surface Area	0.01 m ²
Rudder Chord	0.05 m

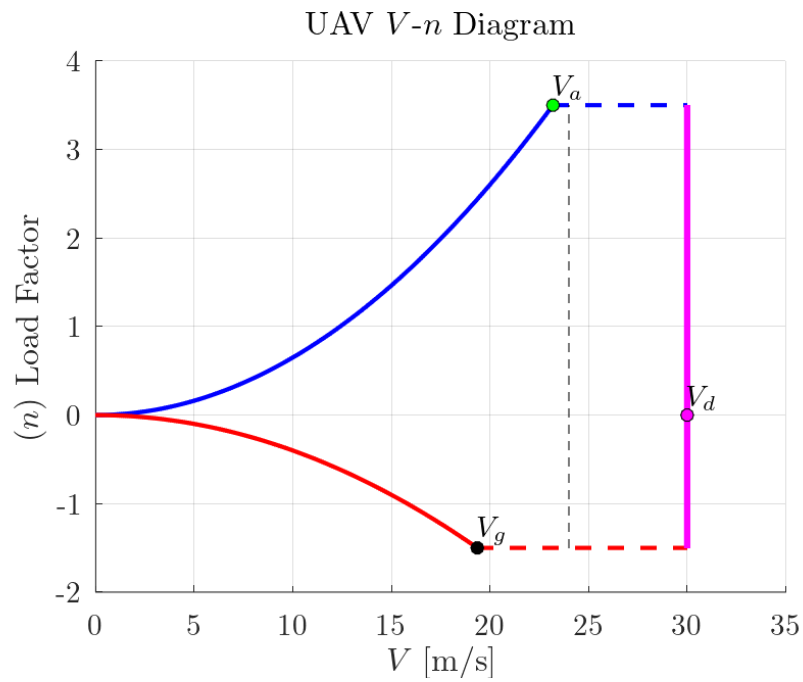
Table 10: Rudder Sizing

Property	Value
Elevator Span	0.77 m
Elevator Surface Area	0.04 m ²
Elevator Chord	0.05 m

Table 11: Elevator Sizing

Locations?

6 V-n diagram



Not enough detail
Expected important velocities,
limits, areas,
max and min load factors,
corner speed, dive speed,

Figure 3: V-n Diagram

3 View CAD model is missing

7 Bill of Materials

A rough estimate of the bill of materials is shown below. This gets updated as design progresses.

7.1 Structural Elements

Component	Rate	Time req. for procure-ment	Vendor	Qty	Price
Carbon Fibre rod, 20mm(OD) 18mm(ID) 500mm(L)	1039	30 days	Robu India	1 (each)	1039
Aluminium Sheets, 0.5mm 5052 H32	345	30 days	Indiamart	10 (kg)	3450
Landing gear wheels (Foam Rubber 90mm)	450	30 days	Flying Machines	3 (each)	1350
Landing gear bearings (4x8x3mm Deep Groove)	1816	10 days	Amazon India	1 (set)	1816
Wheel Axle (4mm x 300mm Stainless Steel Rod)	297	10 days	Amazon India	1 (set)	297
				Total	7952

7.2 Propulsion

Component	Rate	Time req. for procure-ment	Vendor	Qty	Price
T-Motor AM480 900KV	7760	15 days	Robokits India	2 (each)	15520
Propeller (APC 12x6E)	690	25 days	Indiamart	2 (each)	1380
ESC (Hobbywing SKYWALKER-50A-V2)	1970	20 days	Robu India	2 (each)	3940
				Total	20840

7.3 Avionics

Component	Rate	Time req. for procure-ment	Vendor	Qty	Price
Transmitter and Receiver (FS-i6/FS-iA6)	3464	15 days	Robu India	1 (each)	3464
GPS Module (Beitian BN-220)	1867	20 days	Electropi	1 (each)	1867
Video Transmitter (TS835 5.8G)	1685	20 days	Indian Robo Store	1 (each)	1685
Power Distribution Board (PDB-XT60)	373	20 days	Robu India	1 (each)	373
Servo Extension - 90 cm	117	20 days	Robu India	3 (each)	351
Servo Extension - 60 cm	68	20 days	Robokits India	2 (each)	136
Bullet Connector Male-Female pair	25	20 days	Robokits India	4 (each)	100
Servo Extension - 100 cm	128	20 days	Robu India	3 (each)	384
Flight Controller (Pixhawk 2.4.8)	6599	20 days	Robu India	1 (each)	6599
Power Module (Hex Cubepilot Mini)	2659	20 days	Indian Robo Store	1 (each)	2659
Servo Power (Hobbywing UBEC 10A)	4499	20 days	Indian Robo Store	1 (each)	4499
Telemetry Module (3DR 915MHz)	5556	20 days	Robokits India	1 (each)	5556
				Total	27673

7.4 Controls

Component	Rate	Time req. for procurement	Vendor	Qty	Price
Servo Motor (TowerPro MG996R)	343	20 days	Robu India	5 (each)	1715
XT90 Amass Connectors - Pair	108	20 days	Robokits India	2 (each)	216
Control rods (RC Airplane Linkage)	429	10 days	Amazon India	2 (each)	858
				Total	2789

7.5 Payload

Component	Rate	Time req. for procurement	Vendor	Qty	Price
Battery (GenX 22.2V 6S 12000mAh)	12910	15 days	Robokits India	1 (each)	12910
FPV Videography Camera (SKYDROID Mini)	2499	15 days	Indian Robo Store	1 (each)	2499
Parachute (2Kg nylon)	4200	20 days	RcHyderabad	1 (each)	4200
				Total	19609

Grand Total: ₹78863

BOM is not needed in the report

8 Contributions

Table 12: Team Contribution – Week 1

Name	Contribution
DS Vivek	
Mahesh Kumar S	Design Summary
Soumil BM	Design Summary
Manav Agarwal	Bill of Materials
Abhijeet Mangela	Bill of Materials